Total Return Swaps

Seminar Module

conducted by

ACF Consultants Ltd





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Total Return Swaps

- In a Total Return swap there are two counterparties:
  - The Total Return Payer
  - The Total Return Receiver
- Total Return Receiver
  - Receives total rate of return from a reference asset
  - Pays LIBOR plus a spread
- Total Return Payer
  - Pays total rate of return from a reference asset
  - Receives LIBOR plus a spread
- A Total Return swap is an off-balance sheet transaction
- Total Return Receiver therefore enjoys all cash flow benefits from a reference asset…
- …but without actually owning it

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The Total Return

Total Return comprises:

- Regular dividends or interest
- Capital appreciation (or depreciation)

Capital Appreciation = Final Value less Original Value

- In the event of default, “capital appreciation” will reflect the capital loss

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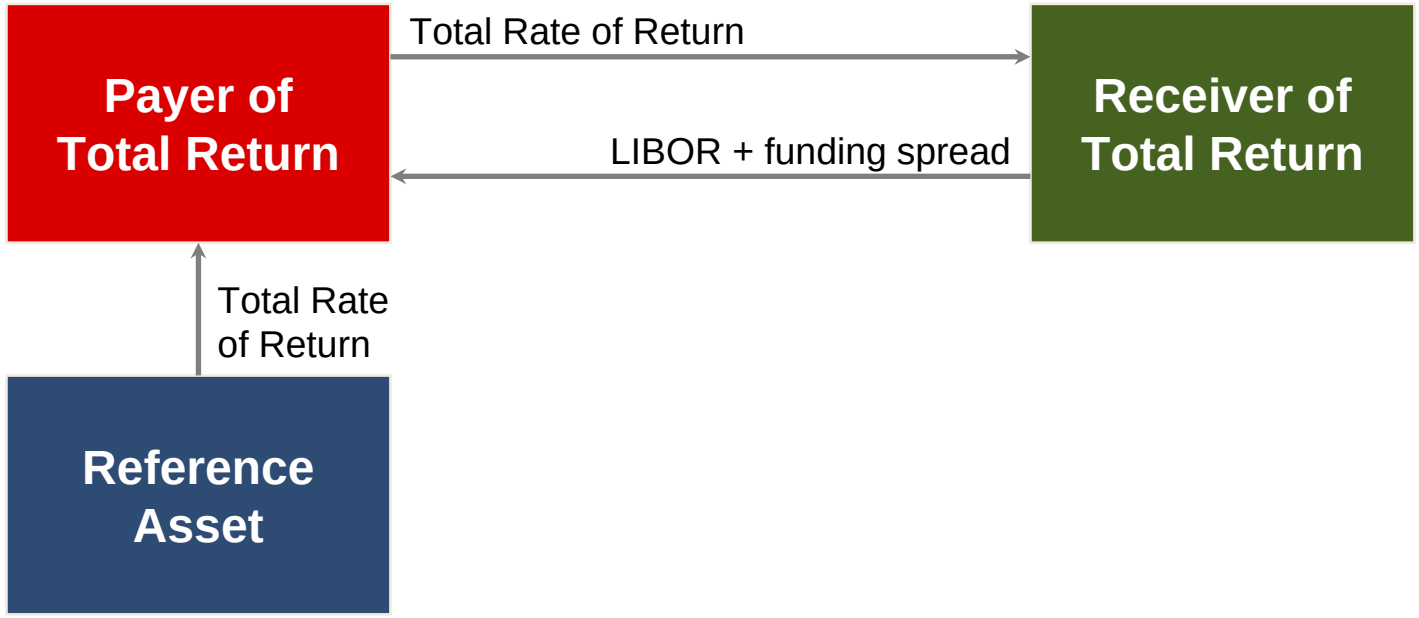
Cash Flows from Total Return Swap

The Total Return Payer:

- Is the legal owner of the reference asset
- Shows the reference asset on their balance sheet
- Is short market risk and credit risk

The Total Return Receiver:

- Does not own the reference asset
- Has an off-balance sheet instrument
- Is long market risk and credit risk

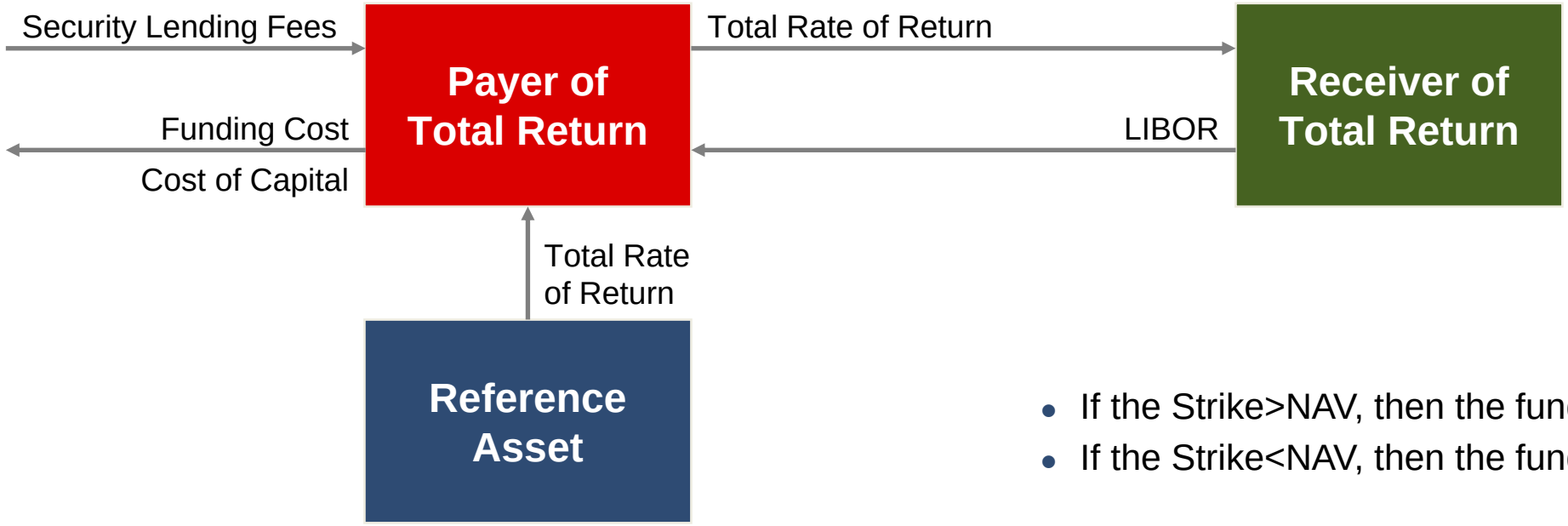


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Cash Flows from Total Return Swap – Practical Features

- The Total Receiver actually pays LIBOR flat
- The Total Return is adjusted by setting the initial *Strike* price slightly different from the initial price (NAV) of the reference asset

- The difference reflects:
  - The TR Payer's actual funding cost
  - Fees the TR Payer earns from lending the reference asset
  - TR Payer's regulatory capital requirements
  - Supply and demand for the reference asset



- If the Strike>NAV, then the funding spread is positive
- If the Strike<NAV, then the funding spread is negative

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Example TRS Term Sheet

Total Return Payer:	Goldman Sachs
Total Return Rcvr:	Deutsche Bank
Reference Asset:	Common Shares
Issuer:	Ford Motor Company
Collateral Type:	Senior unsecured
Initial Share Price:	6.90
Strike Price (P_0):	6.9125
Initial 3m LIBOR:	0.25%
Calculation Amount:	$100,000,000 (fixed)
Calculation Agent:	Total Return Payer

Trade Date:	23rd June 2020
Effective Date:	24th June 2020
Maturity Date:	22nd Sep 2020
Final Accrual Date:	24th Sep 2020
TR Payer pays:	All cash flows from reference asset
TR Receiver pays:	3mth LIBOR A/360
Termination pmnt:	All accrued payments plus $A \times [(P_t / P_0) - 1]$ from TR Payer to Receiver

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Example TRS Calculations

Suppose Ford's share price at maturity was $6.82, and that Ford paid no dividends over the period.

TRS Cash flows are therefore:

- TR Receiver *pays* $1,338,436.22

$$\left(\frac{6.82}{6.9125}-1\right)\times 100{,}000{,}000 = -1{,}338{,}155.52$$

- TR Receiver also pays 63,888,89

$$0.25\% \times 100{,}000{,}000 \times {}^{92}\!/_{360} = 63{,}888.89$$

- TR Payer's implied funding spread

$$\frac{\left(\frac{6.9125}{6.90}-1\right)}{\left(\frac{92}{360}\right)} = 0.71\%$$

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Default Provisions

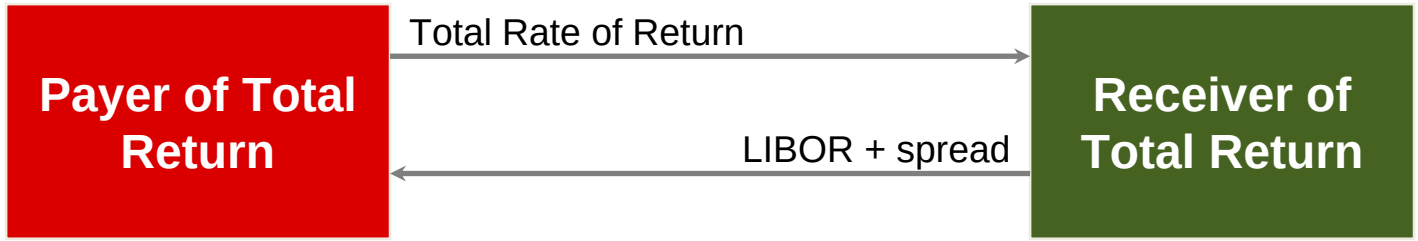
In the event of a default by the issuer of the reference asset:

- Total Return Receiver bears the financial burden of the default
- To compensate for default, the Total Return Receiver must either:
  - Pays Total Return Payer (P_0 - P_d),or
  - Take delivery of the defaulted reference asset and pay the Total Return Payer P_0
  - where:
    - P_0 is the price of the reference asset at the inception of the swap
    - P_d is the price of the reference asset at the time of default
- Total Return swap then terminates

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Collateral

- In the event of default, financial losses from reference asset must be paid:
  - By Total Return Receiver
  - To Total Return Payer
- Total Return Payer therefore has exchanged credit risk of reference asset for credit risk of Total Return Receiver
- TR Payer normally demands collateral from TR Receiver to mitigate credit risk



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Example of Leverage from Total Return Swap

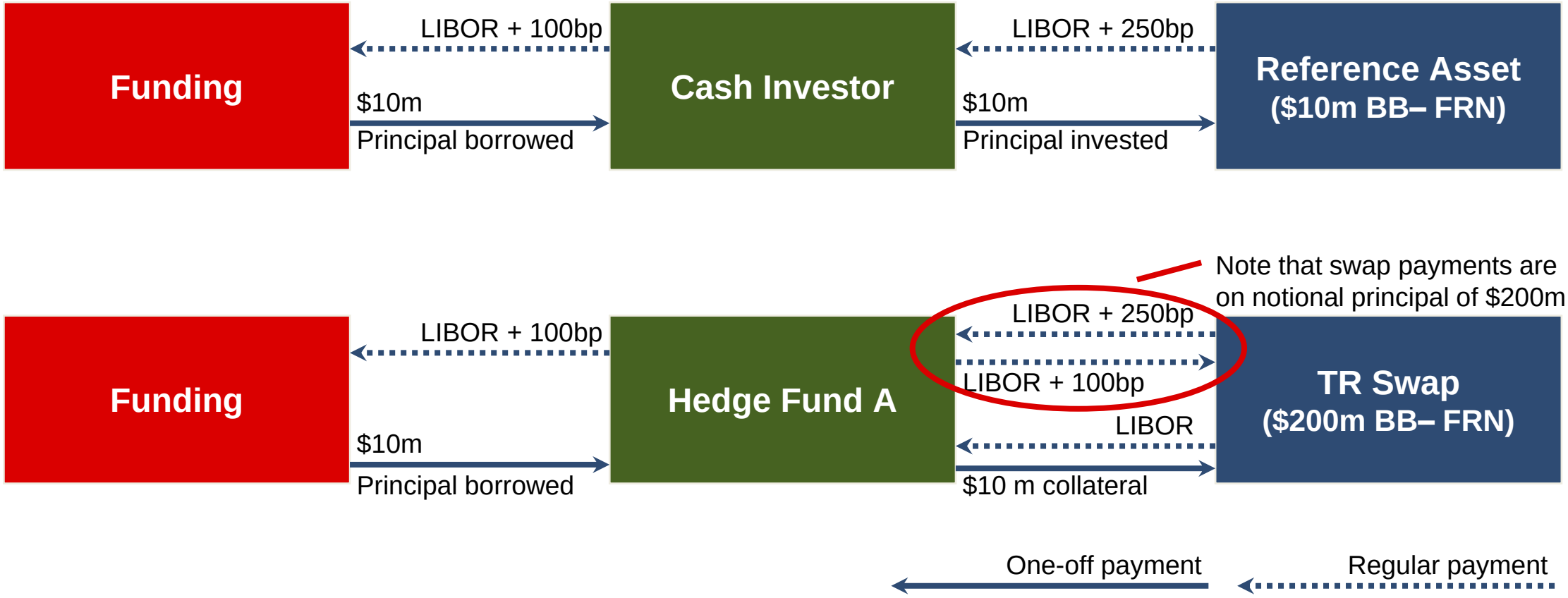
Assumptions:

- Reference asset is a BB- FRN with a coupon of L+250bp
- LIBOR is currently 1.80%
- Hedge funds must pay L+100bp funding cost or as spread on TR swap
- Hedge Fund A must place 5% collateral
- Hedge Fund B must place 10% collateral

	Hedge Fund A	Hedge Fund B	Cash Investor
LIBOR	1.80%	1.80%	
Asset yield	4.30%	4.30%	8.30%
Funding cost	2.80%	2.80%	
Net swap spread	1.50%	1.50%	
Collateral	5%	10%	
Leverage	20 to 1	10 to 1	1 to 1
Levered swap return	30.00%	15.00%	
Net interest on funding	1.00%	1.00%	6.80%
Net return	**29.00%**	**14.00%**	**1.50%**

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Example of Leverage from Total Return Swap



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Motivation of Total Return Payer

- Hedging holdings of assets
- Creating synthetic short positions where this may be difficult in the cash market
- Creating a temporary hedge for an asset expected to suffer a temporary decline
  - ...by entering into Total Return swap with a shorter maturity than the reference asset
- Under some tax regimes, profits and losses can be deferred

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Motivation of Total Return Receiver

- Investors can create new assets with a specific maturity not normally available
- Investors gain efficient off-balance sheet exposure to a defined asset class to which they would not normally have access, e.g.
  - Syndicated loans
  - Emerging market bonds
- Investors may receive a higher return on capital
- Investors may adjust credit exposures in their portfolios
- Investors can reduce administrative costs via off-balance sheet purchases
- Investors can access entire asset classes by receiving the total return on an index
- Leverage, as illustrated earlier

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